



Does relationship lending help firms to ask for credit? European cross-country evidence[☆]

Jérémie Bertrand^a, Marieke Delanghe, Paul-Olivier Klein^{b,*}

^a IESEG School of Management, Univ. Lille, CNRS, UMR 9221 - LEM - Lille Economie Management, F-59000 Lille, France

^b Université Jean Moulin Lyon 3, iaelyon School of Management, UR Magellan, Lyon, France

ARTICLE INFO

Handling Editor: Sushanta Mallick

JEL classification:

G14

G21

D25

D45

Keywords:

Borrower discouragement

Lending technology

Access to credit

Relationship lending

ABSTRACT

Relationship lending is well known to allow SMEs to obtain credit, however one of the most crucial obstacles for these firms is not obtaining credit but applying for credit in the first place. We explore how relationship lending impacts the demand for credit, i.e., borrower's discouragement. Using a European cross-country survey of more than 2300 firms in 2010, we show that firms that rely on relationship lending to get access to credit refrain from applying in the first place. We show that this result is due to a self-rationing mechanism: riskier firms employing relationship lending tend to be more discouraged. Instead, the use of transactional lending reduces discouragement, and this effect does not depend on firms' risk. Our results suggest that firms employing relationship lending know their own likelihood of rejection better and can decide when not to apply.

1. Introduction

A key obstacle for firms in general but small and medium-sized enterprises (SMEs) in particular is limited access to credit, not just through rejected applications but also in the form of refusals to apply in the first place. In the United States, [Levenson and Willard \(2000\)](#) find that borrowers that need credit feel discouraged from applying for it, out of fear that they will be turned down, to such an extent that their numbers are twice as many as the borrowers that get rejected or rationed. That is, while credit rationing is often depicted as the main obstacle to credit (e.g., [Jaffee and Russell, 1976](#); [Stiglitz and Weiss, 1981](#); [Cheng and Yang, 2022](#)), discouragement from applying actually may offer a much more acute barrier.

In discussing the causes of discouragement, the literature has emphasized the characteristics of the firms and the role of information asymmetries (e.g. [Kon and Storey, 2003](#); [Han et al., 2009](#)). However, it considers banks as a homogeneous group that adopts the same approach in lending. [Berger and Udell \(2002, 2006\)](#) stress that banks do not lend money the same way. They propose a classification into two main

groups: relationship and transactional lending. Transactional lending banks rely on hard information (e.g., accounting and historical data) and require few contacts with the firm. Relationship lending instead relies more on soft information, informal and relational insights, and regular contacts. Lending technologies have been extensively studied in the literature: both in terms of their determinants ([Bertrand and Klein, 2021](#)), their impact ([Bharath et al., 2011](#)) or, more recently, their substitutes ([Bertrand and Murro, 2022](#)). In this study, we complement the existing literature on the impact of lending technologies by combining this literature with the literature on discouragement. More precisely, we show that *heterogeneous* lending technologies directly affect the extent of credit discouragement – and reconcile some apparently contradictory results in the literature.

In general, relationship lending reduces information asymmetries between the bank and the firm ([Uchida et al., 2012](#)). Diminished information asymmetries can imply two opposing effects on discouragement. On the one hand, by reducing information asymmetries, relationship lending reduces screening errors, the cost of disclosing information, and the cost of credit ([Bharath et al., 2011](#); [López-Espinosa](#)

[☆] We wish to thank the editor, the associate-editor, the copy editor, two anonymous referees, and seminar and conference participants for helpful comments and conversations. All remaining errors are ours.

* Corresponding author.

E-mail addresses: j.bertrand@ieseg.fr (J. Bertrand), mk.delanghe@gmail.com (M. Delanghe), paul-olivier.klein@univ-lyon3.fr (P.-O. Klein).

<https://doi.org/10.1016/j.econmod.2023.106303>

Received 30 June 2022; Received in revised form 20 February 2023; Accepted 3 April 2023

Available online 18 April 2023

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et al., 2017; Davis and Kim, 2017). These are the main drivers of discouragement based on the model of Kon and Storey (2003). On the other hand, relationship lending also imply that the firm have a closer relationship with their banker and might better know either it is strategic to apply for a credit or not. This would follow the explanation of Han et al. (2009), who attribute a substantial part of discouragement to an efficient self-rationing mechanism, based on rational risk assessments and likelihood of success.

Conversely, the use of transactional lending is likely to entail the opposite effects. Transactional lending relies on hard information and impersonal relationship between the bank and the firm. This translates into more information asymmetries in the decision of the bank to grant a loan. For the firm decision to apply for a credit, this can lead to two opposite phenomena: an anticipation of higher screening cost and difficulty in accessing credit (higher discouragement); or, on the opposite, a lower level of self-rationing (lower discouragement). From this perspective, the type of lending technology is playing a pivotal role in explaining borrowers' discouragement due to its impact on the extent of information asymmetries and the decision from the firm to apply for a credit in the first place and to potentially decide to self-ration.

To determine the role played by lending technologies in explaining patterns in firms' discouragement, we adopt an empirical approach. Measuring discouragement is inherently difficult, due to the lack of information available about firms that decide not to apply for credit. At the same time, measuring the use of a specific lending technology also presents some challenges. To overcome these obstacles, we employ a survey of firms from seven European countries collected by EFIGE and Brunel in 2010, spanning a large, representative sample of European firms of over 2000 firms. The EFIGE survey provides a reliable measure of lending technology based on the type of information demanded by the bank. It has notably been used by Ferri et al. (2019) and Bertrand and Klein (2021). The survey also provides a reliable measure of discouragement, allowing us to know if the firm needed and wanted to apply for credit and if it did apply or not. Furthermore, we are able to merge the answers to the survey with the Amadeus database and retrieve firms' specific financial information. While limited in term of time coverage, the EFIGE survey offers a unique and direct way to measure both discouragement and lending technology.

The results confirm the central influence of heterogeneous lending technologies on borrowers' discouragement. We find that the use of relationship lending exacerbates borrowers' discouragement. The results are economically meaningful: firms that employ relationship lending are 47% more likely to be discouraged than firms employing other types of lending technologies. The effect is the opposite for firms employing transactional lending: they are 45% less likely to be discouraged (both technologies are not perfect substitutes). We explore the potential explanations leading to these results. We find that, in the case of relationship lending discouraged borrowers are essentially risky firms; on the contrary risk does not play a moderating role in the case of transactional lending. This suggests that the type of lending technology modifies the extent of self-rationing by firms. Risky firms that employ relationship lending are able to effectively self-ration based on their risk; risky firms that employ transactional lending do not. In additional results, we also show that discouragement due to relationship lending mainly emerges among firms in countries with higher creditor information, going along the line of a self-rationing explanation. Furthermore, we find that although relationship lending leads to more discouragement for both SMEs and larger firms, the magnitude is lower for SMEs. Both results suggest some form of financial constraint associated with discouragement in the case of relationship lending.

One specific feature of the survey is that it was conducted in 2010, in the aftermath of the financial crisis. From this perspective, our results must also be seen as occurring in a period during which information asymmetries were high and access to credit more difficult for firms. A riskier and opaquer environment is likely to reinforce the polar role of lending technologies in impacting borrowers' discouragement. This is

interesting because part of the literature has pointed to the role played by lending technologies across the business cycle. Bolton et al. (2016) and Berck and al. (2018) have notably documented a reduction in credit rationing associated with relationship lending during a crisis and an economic downturn. We do not possess estimates over the cycle. However, our results suggest that, *even in a downturn*, relationship lending fosters discouragement. This somehow qualifies the view that relationship lending provides easier access to credit during downturns. In fact, part of the reduction in credit rationing observed in previous studies might be due to a higher level of discouragement. We lack the data for a formal demonstration of this effect and limit ourselves to suggesting that future studies on access to credit more frequently include the effect on discouragement, which might qualify the overall picture.

We conduct several robustness tests. Our results rely on the hypothesis that the type of information is at the heart of the lending technology employed. We employ the concept of informational mismatch à la Ferri et al. (2019). Employing situations of mismatch between firms' type of information and the type of lending technology, we confirm that it is the ability of a lending technology to deal with soft information which is the determinant of discouragement, rather than other factors. Our results are also exposed to endogeneity concerns. These are of two sorts. First, firms might select a lending technology based on their likelihood to be discouraged; this is a typical reverse-causality problem. We deal with this potentiality with instrumental variables borrowed from the literature and exogenous to the type of lending technology deployed by the bank. Second, firms might not report being discouraged if their declaration depends on their likelihood to be discouraged; this is a selection bias problem. We accommodate this situation by running selection equations for firms that declare that they did not need to apply for a credit in the first place. In all cases, our results are robust to these endogenous relationships. We further confirm the causality of our mechanism by looking at a subset of financially highly dependent firms and firms with limited funding alternatives. Last, we also employ alternative measurements that confirm our main findings.

Our findings suggest that lending technologies play a pivotal role in firms' discouragement. Our results tilt toward Han et al. (2009) self-rationing mechanism. They show that this self-rationing mainly occurs for the subset of firms that employ relationship lending. These firms seem to be able to secure private information on their odds of success in the case in which they decide to apply for a credit. The use of relationship lending allows them to strategically decide when (not) to apply. Our results confirm the role of information in discouragement (Kon and Storey, 2003). However, they show that information exerts a non-trivial impact: firms are not less discouraged when using relationship lending due to a reduction in information asymmetries; they are more discouraged due to a better knowledge of their chances of success.

These results have important implications for regulators and managers. For policymakers, they reveal that the type of lending technology affects the provision of credit in the market due to their effect on the number of credit applications (and not only the number of rejections). For companies and bank managers, our results uncover an overlooked impact of building long-term banking relationship: while borrowers employing relationship lending might apply less for a credit in the first place, most of this effect comes from a more effective self-rationing behavior, likely to save time and cost in the processing of credit applications.

The remainder of the study is as follow. In Section 2, we review prior literature in detail to establish our hypotheses. Section 3 contains the data and methodology. After we present the main results in Section 4, we explore the detailed roles of risk and alternative sources of funding in Section 5, then offer robustness tests in Section 6. Section 7 concludes.

2. Literature and hypotheses

2.1. Discouragement

The main theoretical framework to describe borrowers' discouragement is the one proposed by [Kon and Storey \(2003\)](#), which highlights a central role of information asymmetries. Information asymmetries also can explain credit rationing (e.g., [Stiglitz and Weiss, 1981](#)), which takes place after the application for credit has reached the bank. But [Kon and Storey \(2003\)](#) offer a novel view by identifying three consequences of information asymmetries in terms of discouraging borrowers even to apply: screening errors by banks that cannot distinguish good and bad borrowers; application costs, which are mostly financial, in-kind, or psychological in nature, related to providing information; and higher interest rates than offered by the market or other sources of funding. This model predicts that borrowers internalize potential screening errors, put them into perspective relative to their own application costs, and compare the offer with what the market provides. Reductions in screening errors or application costs might also reduce discouragement. Available (cheaper) alternative sources of funding reinforce the relative cost though and enhance discouragement.

This straightforward framework underlies most empirical studies of borrowers' discouragement. In line with the information asymmetries suffered by SMEs and their few alternative sources of funding, this research stream tends to focus on discouragement among SMEs. For example, [Levenson and Willard \(2000\)](#) establish that discouraged U.S. firms tend to be smaller and owned by their founders. In international settings, [Chakravarty and Xiang \(2013\)](#) also confirm that discouraged borrowers are smaller and younger, as do Mac an Bhaird, Vidal, and Lucey (2016) in nine European countries. All firms may experience discouragement, but SMEs' characteristics appear to make them especially prone to it. Furthermore, [Brown et al. \(2011\)](#) and Gama, Duarte, and Esperança (2017) show that opaque firms and those more geographically distant from their bank, such that their information asymmetry likely is greater, also experience more discouragement.

An interesting feature of [Kon and Storey's \(2003\)](#) model is that it depicts banks as homogeneous. Choosing one bank over another does not affect the probability of being discouraged. But informal characteristics of the relationship between the borrower and the bank could be important, according to studies such as [Chandler's \(2010\)](#), which shows that greater borrower satisfaction increases discouragement. In addition, [Rostamkalaei et al. \(2020\)](#) find that informal discussions increase discouragement for borrowers that have a good relationship with the bank. This stream of research thus challenges the assumption that the relationship between a borrower and a bank is homogeneous across borrowers, even if it still anticipates that all banks behave homogeneously. We seek to go a step further to generalize the approach by introducing heterogeneity in lending technologies, in line with consistent evidence of their variance.

2.2. The role of lending technologies

Lending technologies are processes put in place by banks when lending to a borrower. [Berger and Udell \(2002\)](#) propose two main types of lending technology, based on the use of soft or hard information and the type of relationship between the bank and the borrower. Transactional lending uses mostly hard accounting and historical information about the firm, during limited interactions. Relationship lending relies mainly on soft, non-recordable information, obtained informally, usually through long-term relationship ties.¹ Predictions tend to imply relationship lending should be highly relevant for SMEs ([Berger and](#)

[Udell, 2006](#)), whereas transactional lending often is assumed to apply more to public companies. But these designations might not be so clear-cut in practice ([Berger and Black, 2011](#)). We propose two channels through which lending technologies might impact borrowers' discouragement: their effect on information asymmetries and their effect on self-rationing.

2.2.1. Lending technology and information asymmetries

Transactional and relational lending technologies reduce information asymmetries to different extents. First, these different lending technologies rely on different screening methods. The very existence of relationship lending is perceived as a way to reduce screening errors and provide credit to firms with less verifiable income ([Boot, 2000](#)). From this perspective, firms engaged in relationships with their bank may be less likely to be deterred by potential screening errors, whereas transactional banks, with their reliance on hard information, might exacerbate screening errors.

Second, heterogeneity in lending technologies affects the application process. Firms engaged in relationship banking already have regular contacts and offer substantial information disclosures, particularly of informal information ([Uchida et al., 2012](#)). The financial costs associated with a loan application, such as due diligence, costs in-kind, or psychological costs, all should decrease, because the relationship already features frequent meetings and shared personal information. This means lower disclosure costs for firms employing relationship lending. Transactional lending demands significant disclosures of new, hard information. This type of information is not necessarily costly to produce for the firm; it however requires an additional effort compared with firms that are already in a long-term relationship with their banker. In a recent paper, [Bertrand and Murro \(2022\)](#) demonstrate that firms that cannot access relationship lending will substitute it with trade credit from their suppliers.

Third, lending technologies influence interest rates charged by banks. The link between lending technologies and cost of credit is not clear cut. In relationship lending research, findings suggest that firms start benefiting from better pricing of their loans two years after the start of the relationship with their bank ([López-Espinosa et al., 2017](#)), especially when they are in dire need of funds ([Bolton et al., 2016](#)) and as they progress along the business cycle ([Beck et al., 2018](#)). Yet, this effect remains uncertain: some studies have documented no decrease in the cost of credit for relationship lending, and even situations of hold-up during which banks charge higher interest rates ([Sharpe, 1990](#); [Steijvers et al., 2010](#)). In any case, discouragement partly stems from premiums applied to the cost of credit, due to information asymmetries, which deter firms from applying for credit. If relationship lending reduces this premium and alters credit conditions ([Bharath et al., 2011](#)), it might reduce borrower discouragement, compared with other lending technologies.²

Heterogeneous lending technology differently impacts information asymmetries between the bank and the firm ([Uchida et al., 2012](#)). This in turn might influence the three factors that explain borrowers' discouragement in [Kon and Storey's \(2003\)](#) model. Up to an extent, compared with transactional lending, relationship lending might alleviate discouragement by reducing screening errors, costs, and interest rates ([Kon and Storey, 2003](#)). This is our first hypothesis.

H1. Relationship lending reduces borrowers' discouragement, while transactional lending relatively increases it.

2.2.2. Lending technology and self-rationing

There is another view though, in line with [Han et al.'s \(2009\)](#)

¹ [Berger and Udell \(2002\)](#) include three subcategories of lending in transactional lending: financial statement lending, asset-based lending and credit scoring.

² Firm risk is likely to play a moderating role in this mechanism. We discuss and focus on the moderating effect of firm's risk in the Additional Results section.

discussion of discouragement and self-rationing. Relationship lending, by tempering information asymmetries between the firm and the bank, should increase the knowledge that the firm has of its odds of success; it learns about its bank and its likelihood of rejecting the credit application. If that likelihood seems high, the firm would experience discouragement. Frequent, informal talks before the application, as described by Rostamkalaei et al. (2020), can further reinforce this mechanism.

Furthermore, relationship lending requires a relationship building process, usually based on mutual trust (e.g., Ennew and Binks, 1996; Uchida et al., 2012). In their paper, Tang et al. (2017) demonstrate that the higher the trust between the firm and the loan officer, the lower the probability to be discouraged, hence trust play a critical role in discouragement. However, if trust is a critical determinant for relationship lending, it's not the only determinant. Relationship lending also rests on the exchange of soft information, allowing the bank to better evaluate the firm and firms to have access to inside knowledge from their bank. If firms engage in transactional lending, they are not committed to a relationship with the banker, nor do they have inside knowledge of their banker's appetite for risk. From this perspective, they might be more prone to submit an application.

If varying information asymmetries associated with different lending technologies lead firms to establish different estimates of its odds of success, relationship lending arguably could exacerbate borrower discouragement, while transactional lending would temper it. Risky firms using relationship lending might know their odds of success and be able to choose strategically not to apply, implying more discouragement. According to this reasoning, risk is not a moderating factor for firms engaged in transactional relationships, because they have no sense of when it is more strategic to self-ration. As Han et al. (2009) assert, self-rationing might explain discouragement, because being discouraged constitutes an efficient self-rationing mechanism. This effect of risk for predicting discouragement also has been confirmed by Chandler (2010) and Gama et al. (2017).

H2. Relationship lending increases borrowers' discouragement due to a self-rationing mechanism, while transactional lending relatively decreases it.

In summary, we find several arguments for why heterogeneity in bank lending technology might help explain borrowers' discouragement. Yet the precise impact of relationship or transactional lending on discouragement is unclear. Diminished information asymmetries can reduce the related costs and thus decrease discouragement (Kon and Storey, 2003), but they also might evoke self-rationing mechanisms and thus reinforce discouragement (Han et al., 2009). With an empirical approach, we seek to determine the impact of heterogeneity in lending technologies on borrowers' discouragement.

To the best of our knowledge, the only paper that studies the impact of banking relationships on discouragement is Chandler's (2010) working paper. They show, through a simple univariate analysis, that discouraged firms have a longer relationship with their bank. Their intuition is that these firms have a better idea of their quality. The goal of our paper is to take this analysis a step further by directly studying the probability of being discouraged as a function of the type of information exchanged. Our study complements the one of Ferrando and Mulier (2022) by employing a cross-countries sample in Europe.

3. Empirical approach

In this section, we present the sources for our data first, then detail the methodology employed to analyze these data. Finally, we provide some descriptive statistics about the sample.

3.1. EFIGE data set

Measuring discouragement can be difficult, because usually there are no records of firms that do not apply for credit. We take advantage of the

EU-EFIGE/Bruegel-UniCredit³ data set (EFIGE data set), which offers a straightforward way to measure firm discouragement, as well as relational and transactional lending – three essential variables, that are usually hard to estimate with other databases. It mostly surveys SMEs, for which discouragement is critical (Chakravarty and Xiang, 2013). This study uses the survey conducted in 2010 among firms in seven European countries (Austria, France, Germany, Hungary, Italy, Spain, and the United Kingdom). This means that it is in the aftermath of the 2008 financial crisis, a situation we discuss when analyzing the results. The sample is representative of the distribution of firms in each country in terms of size and industries. While we are aware that the EFIGE database is limited in term of time coverage, it also offers a unique way to measure discouragement and lending technologies.

The survey covers six firm-relevant topics: structure; workforce; investment, technological innovation, and R&D; internationalization; markets and pricing; and finance. We mostly rely on the finance items, related to the firm's financing needs, behavior in relation to those needs, and relationships with financial partners. To understand the key variables of interest, such as discouragement and relationship lending, we gather answers to specific questions in the survey. We also obtain financial details about the firms by linking the EFIGE database with the Amadeus database, provided by Bureau van Dijk. Thus, we retrieve financial data (balance sheet and income statement) and general firm information.

The sample garnered from these two data sets is restricted, in accordance with our interest is in discouragement, to firms that need credit and turn to bank credit to meet their needs (43% of the total sample).⁴ We integrate the information about these firms available from Amadeus and remove observations with missing data. The final sample is composed of 2388 firms.

3.2. Methodology and variables

Our goal is to investigate whether the lending technology used in the bank–firm relationship is a significant determinant of discouragement probability. Thus, we first measure the use of different types of lending technologies and the extent of discouragement. We take advantage of the EFIGE survey, such that the measure of borrower discouragement is based on answers to questions F13 and F14, reproduced here⁵

F13. During the last year, was the firm willing to increase its borrowing at the same interest rate?

- Yes.
- No.

F14. During the last year, did the firm apply for more credit?

- Yes, applied for it and was successful.
- Yes, applied for it but was not successful.
- No, did not apply for it.

As explained in the data section, we focus solely on firms that need credit, so all firms in our sample answered “Yes” to question F13. As a robustness check, we conduct a two-stage Heckman selection test to ensure the results are robust to a potential selection bias involving firms that do not need credit.

Among the sample, we categorize firms as discouraged if they were

³ The EU-EFIGE/Bruegel-UniCredit dataset is part of the EFIGE project (European Firms in a Global Economy: internal policies for external competitiveness), supported by the Directorate General Research of the European Commission through its 7th Framework Programme and coordinated by Bruegel. For a complete description of the database and project, see Altomonte and Aquilante (2012).

⁴ Focusing on firms that need credit could lead to a selection bias, which we address in the robustness section.

⁵ EFIGE Survey on European Firms in a Global Economy, December 2009, p. 20. Available at https://www.bruegel.org/wp-content/uploads/2015/06/QST_International_Final_.pdf.

Table 1
Lending technology indicator items.

Items	1	0	T/ R
1. Collateral	62.88%	37.12%	T
2. Balance sheet information	88.35%	11.65%	T
3. Interviews with management on firm's policy and prospects	56.64%	43.36%	R
4. Business plan and firms' targets	51.78%	48.22%	
5. Historical records of payments and debt service	43.55%	56.45%	T
6. Brand recognition	15.51%	84.49%	R
7. Other	10.65%	89.35%	

This table contains the seven response options for the question, “Which type of information does the bank normally use/ask for to assess your firm's credit worthiness?” such that each type requires an answer (1 = yes, and 0 = no). It also indicates each item's classification according to the lending technology indicators, as relationship (R) or transactional (T) lending.

willing to increase their borrowing at the same interest rate in the previous year (question F13) but did not apply (question F14).⁶ These two questions reveal if the firm wanted credit and was ready to pay for it based on its risk but eventually decided not to apply for it. This is what we define as a situation of discouragement. The dummy variable *Discouraged* equals 1 for firms that express this evidence of being discouraged and 0 otherwise, which includes firms that answer “Yes” to F13 or either version of “Yes” to F14. In our main analysis, we thus compare firms that need credit but that either do not apply (discouraged) or do apply (not discouraged), regardless of the outcome of the application.

A strand of the literature (e.g., Kon and Storey, 2003) considers that discouragement is strictly defined as firms that would have obtained a credit if they had applied – i.e., creditworthy firms that do not apply for credit. The empirical literature on discouragement essentially assumes that discouraged firms are creditworthy, without further consideration.⁷ In our case, to provide some additional evidence, we perform a simple counterfactual exercise, in which we compare the Z-score of firms that were discouraged with firms that obtained a credit (i.e., not discouraged and not rejected). On average, firms that are categorized as discouraged have a Z-score of 0.819 and firms that obtained a credit a Z-score of 0.921. A *t*-test shows that averages are not statistically different (*t*-statistic = 0.481), suggesting that our sample of discouraged firms are creditworthy.

We derive the *Relationship Lending (RL)* indicator using the methodology suggested by Bartoli et al. (2013) and Ferri et al. (2019). This indicator aims to capture the inclusion of soft information during credit processes, on the basis of the question: “Which type of information does the bank normally use/ask for to assess your firm's credit worthiness?”

⁶ It is worth noting that question F13 does not imply that firms already borrowed but rather that they are willing to increase their leverage.

⁷ Empirically, creditworthiness has always been assumed in the discouragement literature, due to the difficulty of assessing it without a credit application. For instance, Chakravarty and Xiang (2013) classified a firm as discouraged if it answered “yes” to the question: “During the last three years, were there times when the firm needed credit but did not apply because it thought the application would be turned down?” and “zero-times” to the question “how many times did the firm apply for new loan in the past three years?”. There is no specific consideration of creditworthiness. Han et al. (2009) classified discouraged firms as firms that “did not apply because of fear of rejection” (p.416). They do not identify creditworthiness. Chakravarty and Xiang (2013) assume that firms that did not apply to a loan either because (1) the loan procedure was too complicated; (2) interest rates were too high; (3) collateral requirements were too high; and (4) there was corruption in allocation, are discouraged firms, with no explicit consideration of creditworthiness. In a similar vein, Mac an Bhaird et al. (2016) consider that all firms that fear a rejection are discouraged firms. Ferrando and Mulier (2022) do not require discouraged firms to be creditworthy.

(EFIGE survey, question F16). Firms can choose among seven answers (see Table 1). The results reveal that the main indicator used by banks is “Balance sheet information” (item 2), used in 88.35% of cases, followed by “Collateral” (item 1), with 62.88%.

Similar to Ferri et al. (2019), we classify each item in two categories: relationship (R) or transactional (T), as in the last column of Table 1. The classification follows the definition of transactional and relational lending technologies in the literature. For example, “Collateral” and “Balance sheet information” are classified as transactional, because they refer to hard information. In Berger and Udell (2002), it corresponds to financial statement lending and asset-based lending, two subcategories of transactional lending. “Historical records of payment and debt services” can notably be related to credit scoring, another type of transactional lending approach. On the opposite, “Interviews with management on firm's policy and prospects” is a relationship item, based on soft information, similar to “Brand recognition”, which is essentially an intangible asset for which the value depends on the associated soft information. Each item takes the value of 1 if the bank uses it. Then, we construct the RL indicator as follows⁸

$$RL = \frac{\text{item 3} + \text{item 6}}{2}$$

such that the higher the indicator, the more the bank adopts relational lending technologies. We can construct a transactional indicator with the same methodology, so the TR indicator is:

$$TR = \frac{\text{item 1} + \text{item 2} + \text{item 5}}{3}$$

It is important to note that, as in Ferri et al. (2019), we do not attempt to weigh the different items, as such a weighing would inevitably be arbitrary. We control for several firm characteristics that might affect the probability to be discouraged and the lending technology employed (Berger and Udell, 2002): firm risk, based on the Z-score (Altman and Edward, 2013); firm age and size; shareholder structure (*Family firm*); alternative source of funding through belonging to a group (*Group*); legal status as a *Limited Corporation*; the gender of the CEO (*CEO Male*); and several structure ratios, *Leverage*, *Tangibility*, *Liquidity ratio*, *ROA* and *Labor productivity*. Appendix A contains the detailed definitions of these variables and their sources. We also control for the firm's industry with a set of dummy variables. The data set spans seven countries, so national and intranational characteristics could affect the lending technology and the probability of being discouraged, such as due to legal enforcement or the informational environment (Bertrand and Klein, 2021). We control for these characteristics using a set of regional dummy variables (one for each region within a country).

Finally, no theoretical framework offers insights into how to use clustered standard errors by country. However, potential issues could arise if the variance differs by country. Therefore, we test the sensitivity of our results by conducting the estimation with and without clustered standard errors by region within a country.

To link the state of being discouraged with the type of lending technology, we apply the following logit model:

$$\text{Discouraged} = \alpha + \beta \times \text{Lending Technology} + \gamma \times \text{Controls} + \epsilon.$$

In the cross-sectional data set, each firm only enters once. The dependent, dummy variable is *Discouraged*. Then *Lending Technology* is either the RL indicator of relationship lending or the TR indicator of transactional lending. Because RL and TR are not perfect substitutes, we run a separate model that includes them both. In the *Controls* vector of variables, the associated coefficient is γ .

⁸ In the robustness section, we build an alternative indicator of RL that classifies “Business plan and firm's target” as soft information and part of a relationship lending technology.

3.3. Summary statistics

Table A1 in the Appendix contains the descriptive statistics for our sample. We consider both overall statistics and the distinct traits of firms that were discouraged versus those that applied for credit. We test for differences in means between the two groups using a mean difference test (*t*-test).

In the overall sample, 26.4% of firms ($n = 631$) are discouraged; they did not ask for credit even if they needed it. With regard to the relationship lending indicator, about 36% of the firms in the sample use relationship lending technology. Yet we find no statistically significant difference between applicant and discouraged borrowers. Our sample is mainly composed of SMEs, according to both criteria that the European Commission uses to define SMEs (i.e., maximum 250 employees, maximum €50 million of turnover or €43 million on the balance sheet).⁹ Our sample firms employ an average of 82 employees, earn average turnover of €12, 428, 000, and hold average total assets of €11, 865, 000. They have been in business for 22 years on average.

Regardless of whether they applied for credit, the firms are similar in size, age, leverage, R&D, CEO gender, and sectors. However, discouraged borrowers are riskier than applying firms, and they are more likely to be limited corporations. This risk profile aligns with the predictions of Han et al. (2009).

In the correlation matrix (Table A2), the correlation between *Discouraged* and *RL* is positive but not significant. That between *Discouraged* and *TR* instead is negative and strongly significant. This initial finding already suggests a moderating role of lending technology on discouragement. We also note that *Discouraged* is negatively correlated with the Z-score: The higher the Z-score, the lower the firm risk, and the lower the probability of being discouraged. This finding is in line with a positive role of risk in explaining discouragement. Discouraged borrowers often are limited corporations, and they have male CEOs more often, two factors that are positively linked with firm risk (e.g., Faccio et al., 2016).

4. Main results

As detailed previously, we anticipate two opposite effects: Relationship lending might reduce borrowers' discouragement by tempering information asymmetries (Kon and Storey, 2003), or it might reinforce discouragement through a rational self-rationing mechanism (Han et al., 2009). We focus on the impact of using relationship lending, then move on to discuss the impact of transactional lending.

In Table A3 in the Appendix, the first two columns document the impact of the use of relationship lending on discouragement. In each case, we control for the firm's characteristics, as well as industry and country dummies. We present two versions of the model, with and without variance clustered at the country level. The impact of *RL* on *Discouraged* is positive and significant, indicating that firms that use relationship lending tend to be more discouraged than firms that employ other types of lending technologies. The result remains similar even if we cluster the variance by country. The odds ratios for the two models are at the end of the table. Firms that use relationship lending are 47.2% more discouraged on average than firms that do not employ this lending technology.

This result prompts two comments. First, heterogeneity in lending technology affects borrowers' discouragement. The specific relationship that firms build with their bank over time modifies their behavior when they consider applying for credit. Second, using relationship lending leads to more discouragement. This is in line with a self-rationing explanation proposed by Han et al. (2009). Firms that use relationship lending would better know their likelihood of being rejected, so

rationally, they would pre-emptively decide not to apply, even if they do need a credit. Such a behavior would reflect a better knowledge from the firm on its odds of success when it employs relationship lending. To confirm the role of self-rationing as an explanation mechanism, we propose two additional empirical approaches. First, we look at what happens for firms which employ transactional lending; if the specific characteristics of relationship lending drive our results, we should find no or opposite effects for transactional lending. Second, the self-rationing explanation hinges on the logic that discouragement depends on firms' risk. If there is a form of self-rationing for firms employing relationship lending, riskier firms should be more discouraged (Han et al., 2009). We explore this view in the next section.

We first start with transactional lending. If a better knowledge of the likelihood of being rejected explains our results for relationship lending, one would logically expect that this does not happen for firms that employ transactional lending. This follows the opposite reasoning as relationship lending. In their rare contacts with the bank, involving hard information, firms engaged in transactional lending have few opportunities to develop knowledge about the likelihood their application will be rejected. In sort, firms that use transactional lending should then be less discouraged. To test this view, we run the model using *TR* as our main dependent variable. It is worth noting again that *RL* and *TR* are not perfect substitutes, because more than two lending technologies exist (Berger and Udell, 2002). The correlation between *RL* and *TR* is 26.3% (Table A2). As we detail in Table A3, columns 3 and 4, after controlling for firms, industries, and countries, the specifications with and without clustered variance at the country-level both produce a negative and significant coefficient *TR*. The odds ratio shows that firms using transactional lending are 44.7% less discouraged than firms using other lending technologies. Thus, differences in lending technologies have an impact on borrowers' discouragement. In detail, firms that lack any specific relationship with their banks and instead use lending technology based on hard information are less deterred from applying for credit. Relationship lending might increase borrowers' discouragement, by providing an accurate prediction of their likelihood of being rejected, but transactional lending does not provide similar insights and thus reduces discouragement.

Because *RL* and *TR* are not perfect substitutes, firms that rely on relationship lending can seek out transactional loans, and vice versa. Neither variable is a 0/1 dummy variable, so the use of each lending technology could achieve different possible magnitudes. We run the main model with both *RL* and *TR* on the right-hand side, which enables us to address the entire lending technology mix of each firm. In Table A3, columns 5 and 6 contain the results, which remain similar. That is, employing relationship lending increases discouragement; transactional lending reduces it. Both *RL* and *TR* are significant, and the magnitude, though slightly reduced, remains significant.

Two conclusions emerge from this analysis. First, heterogeneity in lending technology affects the extent of borrower discouragement. Second, relationship lending increases discouragement, and transactional lending decreases it. These findings offer support for a rational self-rationing view, according to which relationship lending creates better knowledge about the likelihood of an application being rejected, leading firms not to apply for credit. The use of transactional lending leads to an opposite effect.

It is worth noting that the survey was conducted in 2010, in the aftermath of the 2008 financial crisis. During this period, information asymmetries were higher, and firms were likely to be more deterred to apply for credit in the first place. Our results refer to this specific period and complement the literature on relationship lending over the cycle and during downturns (e.g., Beck et al., 2018; Bolton et al., 2016). While previous studies documented a lower credit rationing for firms employing relationship lending in periods of crises, we find an increase in discouragement. This could suggest that the lower rationing is related to a reduced number of credit applications in the first place. Due to the lack of data over the cycle, we cannot put a formal argument. However,

⁹ https://ec.europa.eu/growth/smes/sme-definition_en. Last accessed December 2020.

it brings some additional insights on the mechanisms underlying the provision of credit through relationship lending and might qualify previous views.

In the next section, we explore three aspects that enrich our findings: the role of risk, which is central to a self-rationing explanation; the role of the country level of creditor information; and the specific case of SMEs versus large firms.

5. Additional results

5.1. Risk

We first explore how risk might help explain our results. Relationship lending increases discouragement, seemingly by reducing information asymmetries and evoking self-rationing mechanisms (Han et al., 2009). To test this claim, we explore the role played by firm risk in our results. We measure risk as the firm Z-score (Altman and Edward, 2013). While employing the Z-score is not without limitations, Altman et al. (2017) show that this measure still has a high accuracy in the detection of default risk. Consequently, it remains widely employed in the literature, notably in the case of SMEs (e.g., McGuinness et al., 2018). We create two subsamples, by dividing the sample at the mean Z-score, so risky firms fall below the mean (lower Z-score) and less risky firms are above it (higher Z-score). We do not use an interaction term on the full sample as our interest lies in the impact of relationship lending for both levels of risk, and not the difference across the two groups.

Columns 1 and 2 of Table A4 report the estimations. If the self-rationing perspective is correct, we should observe a negative effect of relationship lending on discouragement only among risky firms. The results are straightforward: the coefficient of relationship lending is positive and significant only for risky firms. For non-risky firms, the use of relationship lending does not affect the extent of discouragement. Instead, risky firms that use relationship lending appear more discouraged. This suggests that the higher discouragement observed with relationship lending can be attributed to a self-rationing process.

Columns 3 and 4 address transactional lending, again according to the borrower's level of risk, to check for a potential self-rationing mechanism among borrowers that use transactional lending. Because these firms do not have information about the level of risk the bank would be ready to accept, discouragement should be reduced for both risky and less risky firms. This result is exactly what we observe. For both risky and less risky firms, transactional lending reduces discouragement, so the borrower's risk is not a moderating factor if a firm uses transactional lending technology. We thus confirm the opposing impacts of different lending technologies on borrowers' discouragement and that Han et al.'s (2009) predictions hold primarily for relationship lending technology.

5.2. Creditor information

Our results suggest that the effect of lending technology on discouragement depends on the degree of information asymmetries between the firm and the bank, and crucially on the ability of firms to decide to self-ration. Firms in our sample evolve in different national environment across which the extent of information asymmetries varies. This is likely to affect our results. Notably, Mc Namara, O'Donohoe, and Murro (2020) find that SMEs in countries with less sharing of credit information are more likely to be credit rationed. In a general way, nation-wide level of creditor information exerts a substantial impact on firms' access to credit (e.g., Djankov et al., 2007; Cho et al., 2014).

We now explore the role of country-wide degree of creditor information on the link between lending technologies and borrowers' discouragement. Our results suggest that the impact of transactional lending is unlikely to be funnelled through the degree of information asymmetries: firms that employ transactional lending are less discouraged in general. On the contrary, firms that employ relationship lending

seem to be affected by the degree of information asymmetries when deciding to apply or not for a credit. However, this effect is due to a self-rationing mechanism – that is, a better knowledge by the firm on when it is opportune to apply for a credit. The country level of creditor information is likely to affect these results by varying the extent of information asymmetries firms and banks face.

To assess this possibility, we split our sample based on the national level of creditor information. We use the World Bank's Depth of Creditor Information Index. This index captures the extent of information asymmetries faced by creditors based on eight features that public credit registry or private credit bureaus provide. It ranges from 0 to 8. We divide the sample in two categories of countries: on one side we distinguish countries with an index lower or equal to 5 (Spain, Hungary, Austria and Germany), on the other side those with an index higher than 5 (France, Italy, UK). We run our main model for each of these groups and report results in columns 5 to 8 in Table A4.

Consistent with our expectations, lending technologies are differently impacted by the depth of creditor information. The effect of transactional lending on borrowers' discouragement is the same in countries offering high and low creditor information. The coefficient of *TR* is negative and significant in both cases. This confirms that the role of this lending technology on borrowers' discouragement is moderately related with the extent of information asymmetries. On the contrary, the impact of relationship lending on borrowers' discouragement is affected by the country-wide depth of creditor information. Only companies in a strong informational environment that rely on relationship lending are more discouraged. Firms that evolve in countries with limited creditor information are not affected.

The fact that it is companies that evolve in an environment with better creditor information that are more discouraged when employing relationship lending is revealing. It confirms that the impact of relationship lending is not so much in the reduction of information asymmetries – that would reduce rationing, and in turn incentivize firms to apply for a credit in the first place. On the contrary, relationship lending exacerbates discouragement when creditor information is high, that is, in an environment where information asymmetries are already low. It is in such an environment that firms might be better able to decide to apply or not and to strategically self-ration. This corresponds to our finding. This non-trivial role of creditor information confirms our main explanation on the effect of relationship lending on discouragement.

5.3. SMEs v. large firms

Discouragement can affect all firms, but it appears more acute for SMEs (Chakravarty and Xiang, 2013; Levenson and Willard, 2000; Mac an Bhaird et al., 2016), which are smaller, usually not listed, and unlikely to produce detailed financial statements. Thus, they possess less and lower quality information than larger firms (Van Caneghem and Van Campenhout, 2012). Degryse et al. (2012) note that such opacity influences capital structure choices, so in this subsection, we explore whether the selection of a lending technology has the same effect for SMEs and larger firms, in term of discouragement. With this analysis, we also incidentally control for firm size and test if the results hold for both SMEs and large firms.

There are reasons to believe that SMEs might benefit differently from the choice to apply for credit. Their opacity means that SMEs are more likely to benefit from reduced information asymmetries through relationship lending, so the argument that greater information asymmetries evoke more discouragement might be more acute for SMEs than for larger firms. In short, SMEs using relationship lending might be less discouraged than large firms. This prediction does not mean though that they might not be discouraged in general when using relationship lending, because like large firms, SMEs might be more likely to self-ration when they engage in relationship lending.

We explore this mechanism by running the main model with the full sample and adding a dummy variable *SME* that interacts with *RL*

(relationship lending) and *TR* (transactional lending). We do not create subsamples, because our goal is to identify differential effects between SMEs and large firms, and not to compare the effect between the two groups. Thus, *SME* takes a value of 1 if the firm matches the European Commission definition of an SME and 0 otherwise. In our sample, approximately 64% of firms are SMEs.

In Table A4, the coefficient of $RL \times SME$ is negative and significant, indicating that SMEs that use relationship lending are less discouraged than larger firms; however, the sum of the coefficients of *RL* and $RL \times SME$ remains positive and significant and even of similar magnitude compared with the main estimations. Therefore, SMEs may be less discouraged when using relationship lending than larger firms, but they remain discouraged overall. This finding is consistent with the role of opacity for SMEs. Relationship lending helps reduce the high level of information asymmetries for SMEs, compared with large firms, which marginally reduces their discouragement (Kon and Storey, 2003). But the mechanism of self-rationing also remains present, so the overall effect of relationship lending still is an increase in discouragement. Larger firms using relationship lending tend to be more discouraged than SMEs, as reflected in the higher magnitude of the coefficient of *RL*. For transactional lending, we find no difference between SMEs and larger firms, consistent with the limited gains obtained by opaque firms that engage in transactional lending (Ferri and Murro, 2015).¹⁰ As an incidental result, we also note that our overall findings on the role of lending technology on firms' discouragement holds for both SMEs and larger firms.¹¹

To conclude, this section yields three main results. First, the increase in discouragement associated with relationship lending can be attributed to a self-rationing mechanism, because it only affects risky firms. Second, Kon and Storey's (2003) prediction about the impact of alternative sources of funding on borrowers' discouragement depends on lending technologies, such that it holds true only for firms employing relationship lending. Last, among firms using relationship lending, SMEs are less discouraged than larger firms, even if they do remain discouraged overall.

6. Robustness checks

We conduct five robustness tests. First, we rule out alternative interpretation of our findings with a mismatch framework. Second, we propose alternative measures of relationship lending. Then, we control for the potential selection bias relative to firms that need credit, using a two-step Heckman approach. Next, we control for potential endogeneity in the selection of the lending technology by the firm. Last, we conduct an empirical exercise to validate the causal mechanism put forward in the study.

6.1. Ruling out alternative interpretations

We have argued thus far that relationship lending leads to more discouragement by helping the firm learn its likelihood of rejection. But we might record more discouragement resulting from "informal applications," through informal contacts such as email or phone calls. This story is the one offered by Rostamkalaei et al. (2020). In contrast, we argue that it is the lending technology per se, or banks' uses of soft information, that leads to discouragement. From this perspective, informal contact is part of soft information, but it is not an informal application. The risk of confounding these two different interpretations might appear

¹⁰ Among the control variables, it is worth noting that the coefficient of *SME* is not significant. This outcome should not be interpreted on its own though, because the model already includes Size, Age, and Limited Corp., which also measure key characteristics of SMEs.

¹¹ In a robustness test, we run the same estimation removing *Size* from our control variables. Results remain the same and are available upon request.

Table 2
Soft indicator items.

Items	1	0	Soft
1. The bank offers competitive services and funding.	60.70%	39.30%	
2. The bank offers efficient internet services.	21.30%	78.70%	
3. The bank's lending criteria is clear and transparent.	35.09%	64.91%	
4. The bank is conveniently located.	34.35%	65.65%	
5. The bank has an extensive international network.	14.91%	85.09%	
6. The bank offers also a consultancy on strategic financial decisions.	18.30%	81.70%	S
7. The bank has a long-lasting relationship with the firm.	56.00%	44.00%	S
8. The bank has flexible procedures/not constrained by red tape.	28.68%	71.32%	
9. It was the Group's main bank.	9.26%	90.74%	
10. Other	21.54%	78.46%	

This table contains the ten response options for the question, "Which factors are key in the choice of a main bank?" along with the distribution of answers (1 = yes, and 0 = no), and whether each item can be classified as S for Soft information.

high, but we derive a strategy to assess the validity of our interpretation, based on the concept of mismatch or "odd couples" (Ferri and Murro, 2015). The lending technology essentially matches the type of information transmitted by the firm with the correct type of evaluation. If relationship lending per se leads to discouragement, not informal applications, the ability of a bank to deal with soft information is what might lead to more discouragement. In other terms, we should observe that firms that transmit a lot of soft information to a transactional bank that cannot process this information exhibit more discouragement.

We measure mismatch with Ferri and Murro's (2015) method, as adapted by Bertrand and Klein (2021) to EFIGE. Mismatch is defined as a positive difference between a soft information indicator and the *RL* indicator. When this indicator is high, the firm offers mainly soft information, but the bank uses transactional technology and hard information, such that opacity becomes very important. To construct this indicator, we use question F.12 in the EFIGE survey: "Which factors are key in the choice of a main bank?" Firms choose among ten responses, as detailed in Table 2.

With these answers, we can construct the *Soft* indicator as follows:

$$Soft = \frac{item\ 6 + item7}{2}$$

Then the *Mismatch* indicator is defined as (Ferri and Murro, 2015):

$$Mismatch = \{Soft - RL \text{ if } Soft > RL; 0\}$$

Thus, mismatch does not equal 0 for firms that convey soft information but select a transactional bank. It is important to note that the measure of Ferri and Murro (2015) assumes that mismatch only happens for firms with soft information which select a transactional bank. The reason is that hard information can be processed by all types of banks. On the opposite, soft information can only be handled by relational banks. If our interpretation is correct, firms in a mismatch situation should be less discouraged – i.e., it is the capacity of the lending technology to handle soft information that matters. Column 1 of Table A5 contains the results when we run the model with all the control variables and industry and country dummies. We observe a significant and negative effect of *Mismatch* on discouragement, such that firms in a mismatch are less discouraged when considering whether to apply for credit. This finding offers an essential validation of our interpretation: When firms convey soft information, but the bank cannot process it, they are less discouraged. It is thus the treatment of soft information as a whole, not just informal applications, that leads to discouragement. As we predicted, lending technology is a key factor explaining discouragement.

6.2. Alternative Variables

The measure of relationship lending is critical for our study. The EFIGE survey provides a direct way to measure it, based on the firm's relationship with its bank, but we also seek to reproduce our estimations using alternative proxies. First, we consider the length of the relationship; firms with longer lasting relationships with their bank may be more likely to use relationship lending (Degryse and Van Cayseele, 2000). With the log of the length of the relationship between the bank and the firm (López-Espinosa et al., 2017), we generate a dummy variable *Length* (1/0) that equals 1 if the length of the relationship between the firm and its bank is superior or equal to 2 years and 0 otherwise. Second, we address the number of institutions a firm deals with, assuming that firms are less likely to engage in relationship lending if they have multiple banking relationships. The variable *No. Banks* corresponds to the number of banks the firm uses. A dummy variable takes a value of 1 if the firm has fewer than two banks and its main bank is a local one (*Local Bank*), considering that local banks are mainly relational (Stein, 2002). Third, as Bartoli et al. (2013) argue, relationship and transactional lending are complementary, and some firms use both of them (for our sample, their correlation is .26). Thus an increase in relationship lending does not necessarily mean an increase in the relative proportion of relationship lending used by the firm; transactional lending also could increase proportionally. To verify that we are measuring an actual increase in relationship lending, and not simply an increase in the set of information available, we create a ratio equal to *RL/TR*. If this ratio increases, the firm is using more relationship lending than transactional lending to gain access to credit – and this would reflect an increase in the relative use of that lending technology and not simply an increase in the information available. Fourth, we initially used a continuous measure of relationship or transactional lending technology. From our main variables *RL* and *TR*, we create two dummy variables, *RL* (0/1) and *TR* (0/1), equal to 1 if the firm uses that technology and 0 otherwise.

Finally, when constructing our main measure of relationship lending, we followed the literature for the classification of the answers to question F16 of the EFIGE survey. Accordingly, Ferri et al. (2019) do not classify the item “Business plan and firms’ targets” as either relationship or transactional lending. It can be suggested that because a firm's business plan and target are essentially forward-looking, it consists in a form of soft information for the bank. To acknowledge this view, we construct a new variable *Alt. RL* which classifies “Business plan and firm's target” as soft information and part of the deployment of a relationship lending technology. With this classification, each item of *Alt. RL* takes a value of 1/3 and the variable equals 1 if the firm answers yes to all three items: “Interviews with management on firm's policy and prospects”, “Business plan and firms’ targets” and “Brand recognition”.

Table A1 contains the descriptive statistics of the variables. Appendix B provides the correlations of the alternative measures of relationship lending and our main measure. Table A5, columns 2 to 8, lists the regression results. For all the variables but *Log(length)*, employing alternative measures of relationship lending yields similar results. The coefficients of *Length* (0/1), *RL/TR Local Bank*, *RL*(0/1), and *Alt. RL* are positive and significant, whereas the coefficients of *No. Banks* and *TR*(0/1) are negative and significant. Although the logarithm of the length of the relationship is nonsignificant, the finding actually is in line with López-Espinosa et al.'s (2017) finding that relationship lending is not linear with age but rather initiates only two years after the start of the relationship with the bank. Thus our dummy variable *Length* (0/1) is equal to 1 only two years after the start of the relationship and 0 before; it is significant in our model. These measures confirm that an increase in relationship lending leads to an increase in the probability of being discouraged.

6.3. Selection bias

When constructing our sample, we focused on firms that needed

credit (answer “Yes” to question F13), which might lead to selection bias. Firms might answer “No” and assert they do not need credit, even if they do and simply believe they cannot obtain it. From this perspective, stating a need for credit might reflect the likelihood of being discouraged.¹² To address this possibility, we perform a two-stage Heckman regression.¹³ We first model the likelihood of needing credit and calculate the associated lambda from this first-stage regression. In the regression of the likelihood to need credit, firms' characteristics are independent variables. We create the dummy variable *Need*, equal to 1 if the firm needs credit and 0 otherwise, that is, if the firm answered “No” to question F13 in the EFIGE survey. From this first-stage regression, we compute the associated Heckman lambda, then add it to the second-stage regression, which corresponds to our main model. Thus we can take into consideration the selection bias associated with a negative claim about the need for credit. We perform the second-stage regression for both *RL* and *TR*.

The results are in Table A6. In the first-stage regression, the need to obtain credit is significantly linked to the firm's size, age, and leverage. The second-stage regression produces consistent results. Adding the Heckman's lambda from the first-stage regression does not affect the significance or sign of the two variables of interest, such that *RL* remains positive and significant and *TR* negative and significant, confirming that relationship lending is associated with increased discouragement, while transactional lending increases it.

6.4. Reverse causality bias

Our analysis might be marred by a causality problem. A firm might have selected a specific lending technology because of its impact on the likelihood of securing credit. For example, if a firm anticipates some difficulties securing credit, it might have initially sought out a relationship lending bank, to learn in advance if it is worth applying. From this perspective, the lending technology choice might result from a propensity to be discouraged, not the other way around.

We deal with this potential for endogeneity using an instrumental variable regression, with three instrumental variables. First, similar to Ferri and Murro (2015), we use the level of bank mergers and acquisitions (M&A) in the country, which might affect the type of lending technology employed by the bank. If a bank has been merged or acquired, the change of lending technology might be exogenous. We employ the percentage of bank M&As in the country during 2005–2009, using data obtained from the European Central Bank. Second, we turn to a stream of literature that employs country-level aggregates as exogenous instruments (e.g., Keeley, 1990; Laeven and Levine, 2009; Ferri and Murro, 2015). Country-level aggregates might reflect the overall environment firms are facing, so for each firm, they offer an exogenous criterion. We construct two indicators, reflecting the average country levels of our relationship indicator, *RL* (country level), and our transactional indicator, *TR* (country level). These two variables signify the banking and lending environment firms face, which is exogenous to their choice of lending technology.

We run a two-stage regression for both *RL* and *TR*. In the first stage, we obtain predicted values ($\widehat{RL}$ and $\widehat{TR}$). Then in the second stage, we employ the predicted values to explain discouragement. Table A7 contains the results. For relationship lending, column 1 reports the first-stage regression, and column 2 reveals the second stage. Country-level instruments are significant in the first stage, and in the second stage, $\widehat{RL}$ is positive and significant, confirming our main result. For transactional lending, columns 3 and 4 report the first and second stages,

¹² A good explanation of this concept is available from Cole and Sokolyk (2016).

¹³ Alternatively, we also employ avariate probit model (e.g., Freil et al., 2012) that yields similar results.

respectively. In the first stage, *RL* (country level) and *Banks' M&A* are both significant, and then in the second stage, $\widehat{TR}$ is negative and significant, in line with our main results.

Overall, employing a two-stage least squares regression approach to test for potential reverse causality with exogenous instruments confirms our results. The exogenous use of relationship lending increases discouragement, whereas the exogenous use of transactional lending reduces it.

6.5. Confirming a causal mechanism

To confirm the causal explanation in our analysis, that is, relationship lending reduces discouragement via a self-rationing mechanism, we propose to follow the approach adopted in [Ferri et al. \(2019\)](#). Their logic relies on the rationale that, if there is a causal mechanism, it should be especially observed for some specific subsets of firms. In their case, they look at the link between lending technologies and firms' exports. Their hypothesis is that the causal link, and so the causal explanation, should be especially observed for firm firms that are more dependent on external finance due to their production process, and for firms that exports more opaque products. They confirm this view.

We adapt and replicate this approach. We make the hypothesis that our explanation is especially likely to be causal if observed for two subsets of firms: a) firms that are more dependent on external financing due to their production process, and b) firms that have limited access to alternative funding. If there is a causal link between relationship lending and lower discouragement that can be attributed to self-rationing, we should observe that it is *less* the case for both dependent firms (a) and for firms that have limited access to alternative funding (b). We propose one measure of financial dependence and two measures of financial alternatives.

We measure a firm's degree of funding dependence based on the answers to question F3 of the EFIGE survey (reproduced in Appendix C). The variable *High Dependence* equals to 1 if the firm reports being "extremely dependent" on external funding and zero otherwise. We expect highly dependent firms to be less able to self-ration, even when using relationship lending. Hence, if our self-rationing interpretation is accurate, they should be less discouraged than other firms. The first column of [Table A7](#) reports the results, employing an interaction of *High Dependence* with *RL*. Consistent with our predictions, the coefficient *High Dependence* $\times$ *RL* is negative and significant. More financially dependent firms are less discouraged when employing relationship lending, compared with less dependent firms. This gives more validity to self-rationing as an explanation for our results.

Next, we propose two measures to assess a firm's availability of alternative funding to bank debt. First, we follow the literature showing that groups can provide funding to subsidiaries and ease financial dependence to external funding (e.g., [Dewaelheyns and Van Hulle, 2010](#)). The variable *Group* equals to 1 if the firm belongs to a group, zero otherwise. Second, we consider alternative to bank debt more broadly. We construct the variable *Alternative*, an indicator based on answers to question F1 of the survey. We classify a firm as having many alternatives if its debt structure is composed of at least four sources of funding, and bank debt represents a maximum of 25% of the overall debt structure.¹⁴ This measure reflects whether the firm frequently uses multiple sources of financing and can easily switch among them.

For these two measures, we expect firms that benefit from more potential alternative sources of funding to be more discouraged. This is because these firms will have the choice to secure their funding from another source if they anticipate being rationed (as in [Kon and Storey \(2003\)](#) model). Columns 2 and 3 of [table A7](#) report the results. Both

Group $\times$ *RL* and *Alternative* $\times$ *RL* are positive and significant. This is consistent with our predictions and supports the view that our results capture a causal self-rationing mechanism for firms employing relationship lending.

7. Conclusions

Prior literature on borrowers' discouragement assumes homogeneity in banks' behaviors (e.g., [Kon and Storey, 2003](#)). We challenge this hypothesis by introducing the possibility of heterogeneity in the lending technologies employed by banks, in line with evidence of their importance (e.g., [Berger and Udell, 2002](#)). But we also confronted two opposite predictions, justifying an empirical approach: Relationship lending could reduce discouragement by alleviating the information asymmetries between the borrower and the bank ([Kon and Storey, 2003](#)), or it could reinforce discouragement through a self-rationing mechanism when the firm knows its likelihood of being rejected ([Han et al., 2009](#)). Our findings establish a conclusive answer and shed new light on the associated mechanisms.

Heterogeneity in lending technologies matters. It exerts an influence on the likelihood that a firm becomes discouraged. Relationship lending technologies increase discouragement. Transactional lending technologies reduce discouragement. In line with a self-rationing explanation, risk also plays a central role, such that riskier firms are more discouraged when they use relationship lending, but less risky firms are not. Firms that lack alternative sources of funding are not discouraged by any lending technology; they apply even when they use relationship lending. This result holds for both SMEs and larger firms, even if, compared with large firms, SMEs that engage in relationship lending benefit from a relative reduction in the magnitude of their discouragement.

With these findings, we also clarify that extant, sometimes conflicting results depend on lending technologies. Firms are not rationed when they work with transactional banks; self-rationing only arises among firms that have relationships with their banks, and alternative sources of funding only make a difference in relationship lending settings. Thus, heterogeneity in lending technology is a central component of borrowers' discouragement. With this insight, our findings also extend previous studies that note the role of relationship lending in credit rationing, because we show that its effect emerges even before this step, in the actual decision to apply for a credit. Last, incidentally, our results pertain to the period that followed the 2008 financial crisis. While our dataset prevents us from formally obtaining findings over the business cycle, they suggest that firms employing relationship lending are discouraged even during an economic downturn; this somehow qualifies previous findings on the role of relationship lending in ensuring access to credit over the cycle. Firms might also be discouraged in the first place.

These findings have important practical implications. For the regulator, they mean that different types of banks do not offer access to credit in the same way, starting even at the moment that firms decide whether to apply. This stage is critical for understanding the different roles of banks in providing funding to the economy. In terms of managerial implications, these results offer advice for companies in general and SMEs specifically. They highlight a strategic advantage of relationship lending that has not been identified previously. Constructing a long-term relationship with the bank can enable companies to predict the odds that their application will be granted. Building a long-term relationship with a bank for this end also would allow companies to save time in the credit process and better plan their funding strategies. If a firm has doubts about its borrowing capacity, it could develop a banking relationship with the long-term goal of saving time in later credit-granting efforts.

¹⁴ Adopting different thresholds to define the presence of alternative sources of funding leads to consistent results.

Table A1
Descriptive Statistics

	Full Sample		Mean Difference Test		
	Mean	Std. Dev.	Not Discouraged	Discouraged	Difference
Dependent variable					
Discouraged	0.264	0.441	0	1	
Independent variables					
<i>Lending Technology</i>					
RL	0.361	0.337	0.357	0.370	−0.013
TR	0.649	0.285	0.665	0.606	0.059***
Mismatch	0.467	0.403	0.482	0.421	0.061***
<i>Control variables</i>					
Z-score	1.089	0.900	1.086	0.819	0.267*
Log (Size)	3.629	0.933	3.634	3.617	0.016
Log (Firm Age)	2.872	0.749	2.864	2.893	−0.029
Leverage	0.224	0.208	0.226	0.220	0.006
Group	0.183	0.387	0.182	0.186	−0.004
Tangibility	0.296	0.209	0.296	0.296	0.000
Family firm	0.726	0.446	0.731	0.703	0.028
Liquidity ratio	0.893	0.668	0.852	1.055	−0.203***
ROA	0.035	0.124	0.039	0.021	0.018*
Labor productivity	4.547	0.872	4.536	4.590	−0.054
Limited Corp.	0.790	0.407	0.766	0.857	−0.091***
CEO Male	0.924	0.265	0.918	0.940	−0.022
Manufacturing	0.881	0.324	0.878	0.891	−0.013
Energy Production	0.014	0.117	0.014	0.014	−0.001
Water Production	0.084	0.278	0.086	0.079	0.007
Robustness variables					
Log (Length)	2.561	0.838	2.545	2.605	−0.061
Length (0/1)	0.977	0.151	0.977	0.975	0.003
No. Banks	4.189	4.358	4.631	2.957	1.674***
RL/TR	0.503	0.680	0.502	0.506	−0.004
Local Bank	0.788	0.409	0.766	0.849	−0.083***
Alt. RL	0.356	0.107	−0.360	−0.343	−0.016
SME	0.635	0.482	0.699	0.456	0.243***
Need (*)	0.619	0.485			
Observations	2388 (* 5923)		1757	631	

This table contains descriptive statistics for the dependent variable Discouraged, the Relationship Lending variable (RL), and the control variables related to firm characteristics. The first columns display the summary statistics for the full sample; the later columns display the summary statistics for firms that apply or are discouraged. The last column displays the mean test by accounting practice, where * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A2
Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
(1) Discouraged	1.000														
(2) RL	0.017	1.000													
(3) TR	−0.091***	0.263***	1.000												
(4) Z-score	−0.180***	−0.011	−0.025	1.000											
(5) Log (Size)	−0.008	0.194***	−0.037*	−0.011	1.000										
(6) Log (Firm Age)	0.017	0.025	−0.052**	−0.052**	0.103***	1.000									
(7) Leverage	−0.013	0.080***	0.106***	0.035*	−0.020	−0.009	1.000								
(8) Limited Corp.	0.099***	−0.123***	0.086***	−0.137***	−0.203***	−0.142***	0.132***	1.000							
(9) CEO Male	0.036*	0.029	0.001	0.002	0.087***	0.007	−0.014	−0.039*	1.000						
(10) Group	0.005	−0.084***	−0.120***	−0.013	0.412***	−0.067**	−0.083***	−0.134***	0.101***	1					
(11) Tangibility	0	0.062**	0.054**	−0.286***	0.028	0.028	0.341***	0.060**	−0.046*	−0.036	1				
(12) Liquidity Ratio	0.122***	−0.067**	0.014	0.056**	−0.001	0.029	−0.019	−0.065**	0.016	0.026	−0.142***	1			
(13) Family Firm	−0.025	0.043	0.089***	−0.04	−0.097***	0.059**	−0.002	0.104***	−0.075***	−0.155***	0.015	−0.008	1		
(14) ROA	0.058**	−0.043	0.001	0.015	0.015	0.002	0.001	0.006	0.01	−0.062**	0.009	0.122***	0.065**	1	
(15) Labor Productivity	0.025	−0.050*	−0.023	0.027	0.028	0.094***	−0.114***	−0.01	0.053*	0.067**	−0.156***	0.01	−0.029	0.240***	1

This table displays the correlation matrix between the dependent variable (*Discouraged*) and all other variables. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A3
Discouraged and Lending Technologies – Main Estimation

	(1)	(2)	(3)	(4)	(5)	(6)
Discouraged						
RL	0.155*** (0.008)	0.155* (0.099)			0.075** (0.029)	0.075*** (0.003)
TR			−0.374** (0.019)	−0.374*** (0.000)	−0.356** (0.028)	−0.356*** (0.001)
Group	−0.007 (0.954)	−0.007 (0.966)	−0.014 (0.906)	−0.014 (0.937)	−0.015 (0.905)	−0.015 (0.935)
Tangibility	0.175 (0.445)	0.175 (0.289)	0.181 (0.434)	0.181 (0.266)	0.177 (0.443)	0.177 (0.279)
Liquidity Ratio	0.201*** (0.002)	0.201*** (0.003)	0.198*** (0.002)	0.198*** (0.006)	0.196*** (0.003)	0.196*** (0.006)
Family Firm	−0.006 (0.949)	−0.006 (0.940)	0.001 (0.989)	0.001 (0.987)	−0.000 (0.996)	−0.000 (0.996)
ROA	0.605 (0.111)	0.605 (0.147)	0.566 (0.141)	0.566 (0.228)	0.567 (0.139)	0.567 (0.222)
Labor Productivity	0.027 (0.612)	0.027 (0.384)	0.025 (0.640)	0.025 (0.421)	0.026 (0.621)	0.026 (0.363)
Z-score	−0.031 (0.671)	−0.031 (0.779)	−0.029 (0.695)	−0.029 (0.791)	−0.030 (0.681)	−0.030 (0.782)
Log (Size)	−0.091 (0.114)	−0.091*** (0.000)	−0.112* (0.051)	−0.112*** (0.000)	−0.106* (0.069)	−0.106*** (0.000)
Log (Firm Age)	0.073 (0.256)	0.073 (0.406)	0.067 (0.299)	0.067 (0.440)	0.067 (0.299)	0.067 (0.439)
Leverage	0.244 (0.423)	0.244 (0.282)	0.258 (0.394)	0.258 (0.212)	0.256 (0.397)	0.256 (0.203)
Limited Corp.	0.265* (0.055)	0.265*** (0.001)	0.266* (0.055)	0.266*** (0.001)	0.264* (0.056)	0.264*** (0.001)
CEO Male	0.024 (0.869)	0.024 (0.805)	0.025 (0.865)	0.025 (0.808)	0.023 (0.871)	0.023 (0.820)
Industry Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Region Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Region Cluster		Yes		Yes		Yes
Constant	−1.875** (0.010)	−1.875*** (0.000)	−1.530** (0.041)	−1.530*** (0.002)	−1.543** (0.039)	−1.543*** (0.002)
Observations	2388	2388	2388	2388	2388	2388
Pseudo R ²	0.111	0.111	0.115	0.115	0.115	0.115
OR RL	1.382	1.382			1.282	1.282
OR TR			0.413	0.413	0.548	0.548

These regressions show the impact of lending technologies (RL and TR) on the probability of being discouraged, using logit regressions. We control for firm characteristics and country dummies. Models (1) and (2) display the results for the RL analysis, models (3) and (4) for the TR analysis, and models (5) and (6) for RL and TR together. In models (2), (4), and (6), standard errors are clustered by country. The last two lines display the odds ratios for RL and TR. The regressions are robust to heteroscedasticity. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ (p -values are in brackets).

Table A4
Additional Results.

	Risk				Creditor Information environment				SMEs v. Large Firms	
	Risky	Less Risky	Risky	Less Risky	Low	High	Low	High		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Discouraged		Discouraged		Discouraged		Discouraged		Discouraged	
RL	0.103*** (0.006)	0.089 (0.501)			0.086 (0.645)	0.174*** (0.002)			0.374 (0.283)	
TR			−0.453** (0.034)	−0.417** (0.017)			−0.522*** (0.002)	−0.315** (0.012)		−0.489** (0.012)
RL × SME									−0.658*** (0.006)	
TR × SME										−0.125 (0.670)
SME									−0.795** (0.026)	−0.496 (0.227)
Control variables	All	All	All	All	All	All	All	All	All	All
Region Cluster	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−5.913*** (0.000)	−1.600* (0.067)	−5.377*** (0.000)	−1.279 (0.123)	−2.516 (0.128)	−0.594*** (0.001)	−2.193 (0.216)	−0.382*** (0.000)	−0.409 (0.494)	−0.160 (0.773)
Observations	1095	1293	1095	1293	1484	904	1484	904	2388	2388
Pseudo R ²	0.107	0.136	0.112	0.141	0.171	0.072	0.177	0.074	0.065	0.071

These regressions show the impact of lending technologies on the probability of being discouraged using logit regressions. Models (2) to (4) split the sample based on risk (risky firms have a Z-score below the sample mean); models (5) to (8) split the sample based on the level of creditor information environment (low is defined as an index below or equal to 5, high above 5); models (9) and (10) employ the full sample and explore the existence of distinct impacts of lending technologies for SMEs versus larger firms ($SME = 1$ if the firm meets the European Commission definition of SMEs). We control for firm characteristics and country dummies. Standard errors are clustered by country. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ (p -value are in brackets).

Table A5
Robustness Tests: Alternative Variables

	Discouraged								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mismatch	−0.333** (0.011)								
Log (Length) Length (0/1)		0.032 (0.027)	0.323* (0.195)						
No.Banks				−0.018*** (0.006)					
RL/TR					0.084** (0.034)				
Local Bank						0.142** (0.057)			
RL (0/1)							0.254** (0.015)		
TR (0/1)								−0.242** (0.024)	
Alt. RL									0.402*** (0.003)
Control variables	All	All	All	All	All	All	All	All	All
Region Cluster	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−1.567*** (0.004)	−1.853*** (0.000)	−2.745*** (0.000)	−1.929*** (0.000)	−1.876*** (0.000)	−1.652*** (0.003)	−1.876*** (0.000)	−1.812*** (0.000)	−1.829*** (0.000)
Observations	2388	2388	2388	2388	2388	2388	2388	2388	2388
Pseudo R ²	0.118	0.111	0.115	0.111	0.111	0.113	0.111	0.111	0.116

These regressions report results employing alternative variables. The dependent variable is *Discouraged*. The first column is a logit regression with *Mismatch* as the main dependent variable. Columns 2–9 are logit regressions that show the impact of lending technologies on the probability of being discouraged, using alternative measures of the lending technology. Variables are defined in Appendix A. Standard errors are clustered by country. The regressions are robust to heteroscedasticity. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ (p -value are in brackets).

Table A6

Robustness Tests: Self-Selection Model & IV Analysis

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Self-Selection Model			IV			
	Discouraged			First Stage	Second Stage	First Stage	Second Stage
				RL	Discouraged	TR	Discouraged
				0.090**		−0.026**	
				−0.045		−0.031	
				0.003*		0.019	
				−0.088		−0.639	
				0.086		0.555*** (0.000)	
				−0.394			
RL	0.148** (0.012)		0.066** (0.043)		2.967*** (0.000)		
TR		−0.718*** (0.000)	−0.358*** (0.001)				−3.612*** (0.000)
λ	0.684 (0.597)	0.161 (0.842)	0.761 (0.542)				
Control variables	All	All	All	All	All	All	All
Region Cluster	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.299** (0.023)	−0.593 (0.283)	−0.662 (0.297)	0.268** −0.03	−1.480*** (0.000)	0.856*** (0.000)	−3.189*** (0.000)
Observations	5923	2388	2388	2388	2388	2388	2388
Pseudo R ²	0.040	0.157	0.163	All	All	All	All

These regressions show the impact of the lending technology on the probability of being discouraged, controlling for self-selection bias using the Inverse Mills ratio (λ) and instrumental-variable analysis. Columns 1, 2 and 3 correspond to the second stage of our Heckman model, where we include in our main estimation the Heckman λ . Column 4 (6) corresponds to the first stage of our instrumental variable analysis for RL (TR), and column 5 (7) corresponds to the second stage, with Discouraged as the dependent variable. The regressions are robust to heteroscedasticity. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ (p -value are in brackets).

Table A7

Robustness Tests: Confirming Causality.

	(1)	(2)	(3)
Discouraged			
RL	0.033 (0.787)	0.221** (0.012)	0.142 (0.106)
High Dependence	−0.030 (0.105)		
High Dependence x RL	−0.280* (0.075)		
Group		−0.145 (0.449)	
Group x RL		0.368** (0.019)	
Alternative			0.469*** (0.003)
Alternative x RL			0.186** (0.016)
Control variables	All	All	All
Region Cluster	Yes	Yes	Yes
Constant	−1.819*** (0.000)	−1.835*** (0.000)	−1.919*** (0.000)
Observations	2388	2388	2388
Pseudo R ²	0.113	0.112	0.116

These regressions show the impact of the relationship lending technology on the probability of being discouraged, interacting our *RL* indicator with several variables: *High Dependence* in column 1, *Group* in column 2, and *Alternative* in column 3. The regressions are robust to heteroscedasticity. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$ (p -value are in brackets).

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare that they have no known competing financial

Appendix A. Variable Definitions

Variable	Description	Source
Dependent variable		
Discouraged	= 1 if the firm was willing to increase its borrowing at the same interest rate during the last year (question F13) but did not apply for it (question F14)	EFIGE Database
Lending Technology		
RL	Relationship lending indicator based on question F16 in EFIGE. Average value of dummies equals 1 if the firm answers yes to the two items: “Interviews with management on firm’s policy and prospects” and “Brand recognition”	EFIGE Database
TR	Transactional lending indicator based on question F16 in EFIGE. Average value of dummies equals 1 if the firm answers yes to the items: “Collateral,” “Balance sheet,” and “Historical records of payments and debt service”	EFIGE Database
Mismatch	Equal to the positive difference between Soft index of the firm and RL index.	EFIGE Database

(continued on next page)

(continued)

Variable	Description	Source
Control variables		
Z-score	Altman and Edward, 2013 Z-score	Amadeus Database
Log (Size)	Natural logarithm of the number of employees in the firm.	Amadeus Database
Log (Firm Age)	Natural logarithm of the firm age, in years.	Amadeus Database
Leverage	The ratio of firm's total loans to total assets at the end of 2008.	Amadeus Database
Group	= 1 if the firm belongs to a group; 0 otherwise	Amadeus Database
Tangibility	Ratio of net property, plant, and equipment to assets	Amadeus Database
Liquidity ratio	Ratio of current assets to current liabilities	Amadeus Database
Family firm	= 1 if the firm is a family firm; 0 otherwise	Amadeus Database
ROA	Ratio of profit to assets	Amadeus Database
Labor productivity	Natural logarithm of the ratio sales to number of employees	Amadeus Database
Limited Corp.	= 1 if the firm is a limited corporation; 0 otherwise.	Amadeus Database
CEO Male	= 1 if the CEO is a male; 0 otherwise.	Amadeus Database
Industry	Set of dummies, one for each industry (Manufacturing, Energy Production and Water Production)	Amadeus Database
Region	Set of dummies, one for each region within a country (Austria, France, Germany, Hungary, Italy, Spain, United Kingdom)	Amadeus Database
Robustness		
Log (Length)	Natural logarithm of 1 + Length of the relationship between the firm and the bank (in years)	EFIGE Database
Length (0/1)	= 1 if the length of the relationship between the firm and its bank is inferior or equal to 2 years and 0 if longer.	EFIGE Database
RL (0/1)	= 1 if the relationship lending indicator exceeds zero, so there exists a minimum level of relationship lending; 0 otherwise.	EFIGE Database
TR (0/1)	= 1 if the transactional lending indicator exceeds zero, so there exists a minimum level of transactional lending; 0 otherwise.	EFIGE Database
No.Banks	The number of credit institutions the firm deals with.	EFIGE Database
RL/TR	Ratio of RL divided by TR	EFIGE Database
Local Bank	= 1 if the firm no more than 2 banks and if its main bank is local	EFIGE Database
Alt. RL	Relationship lending indicator based on question F16 in EFIGE. Average value of dummies equals 1 if the firm answers yes to the three items: "Interviews with management on firm's policy and prospects", "Business plan and firms' targets" and "Brand recognition"	EFIGE Database
SME	= 1 if the firm is a SME according to European Commission criteria (less than 250 employees, turnover less than €50 m, total assets less than €43 m)	Amadeus Database
High Dependence	= 1 if the firm declares that its industry is dependend or extremely dependent on external financing.	EFIGE Database
Alternative	= 1 if the debt structure composed of at least four sources of funding, and bank debt represents a maximum of 25% of the overall debt structure.	EFIGE Database
Need	= 1 if the firm needs credit and 0 otherwise, such that the firm has answered "No" to question F13 of the EFIGE survey	EFIGE Database
Instrumental variables		
RL (Country level)	Average value of relationship lending technology for the firms in the country.	EFIGE Database
TR (Country level)	Average value of transactional lending technology for the firms in the country.	EFIGE Database
Banks' M&A	Percentage of bank merger and acquisitions in the country during 2005–2009	ECB

Appendix B. Correlation Matrix with Robustness Variables

	Discouraged	RL	TR	Mismatch	RL (0/1)	Log (Length)	Length (0/1)	No.Banks	RL/TR	Main Bank
Discouraged	1.000									
RL	0.017	1.000								
TR	−0.091***	0.263***	1.000							
Mismatch	−0.035	0.344***	−0.003	1.000						
RL (0/1)	−0.020	−0.884***	−0.230***	−0.170***	1.000					
Log (Length)	0.032	0.017	−0.040**	0.122***	0.005	1.000				
Length (0/1)	−0.008	−0.031	−0.016	0.085***	0.038*	0.345***	1.000			
No.Banks	−0.241***	−0.046**	−0.027	0.061***	0.035*	−0.041**	0.042**	1.000		
RL/TR	−0.003	−0.718***	−0.104***	−0.202***	0.663***	0.001	0.020	−0.017	1.000	
Local Bank	0.090***	0.091***	0.040**	0.017	−0.094***	0.032	−0.019	−0.175***	−0.076***	1.000

This table displays the correlation matrix for the dependent variable *Discouraged*, lending technology variables, and all the robustness variables. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Appendix C. EFIGE questions

F1. What is the overall distribution of your firm's debt structure in percentage terms?

- Short term bank debt (up to 12 months)	%
- Medium to long term bank debt (12 months and over)	%
- Short-term securities	%
- Medium and long-term securities	%
- Other financial instruments	%

F3. In the industry your firm works, how dependant are companies on external financing? To give your answer please use a score from 1 (not dependent at all) to 5 (Extremely dependent)

- 1 not dependent at all
- 2
- 3
- 4
- 5 extremely dependent

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